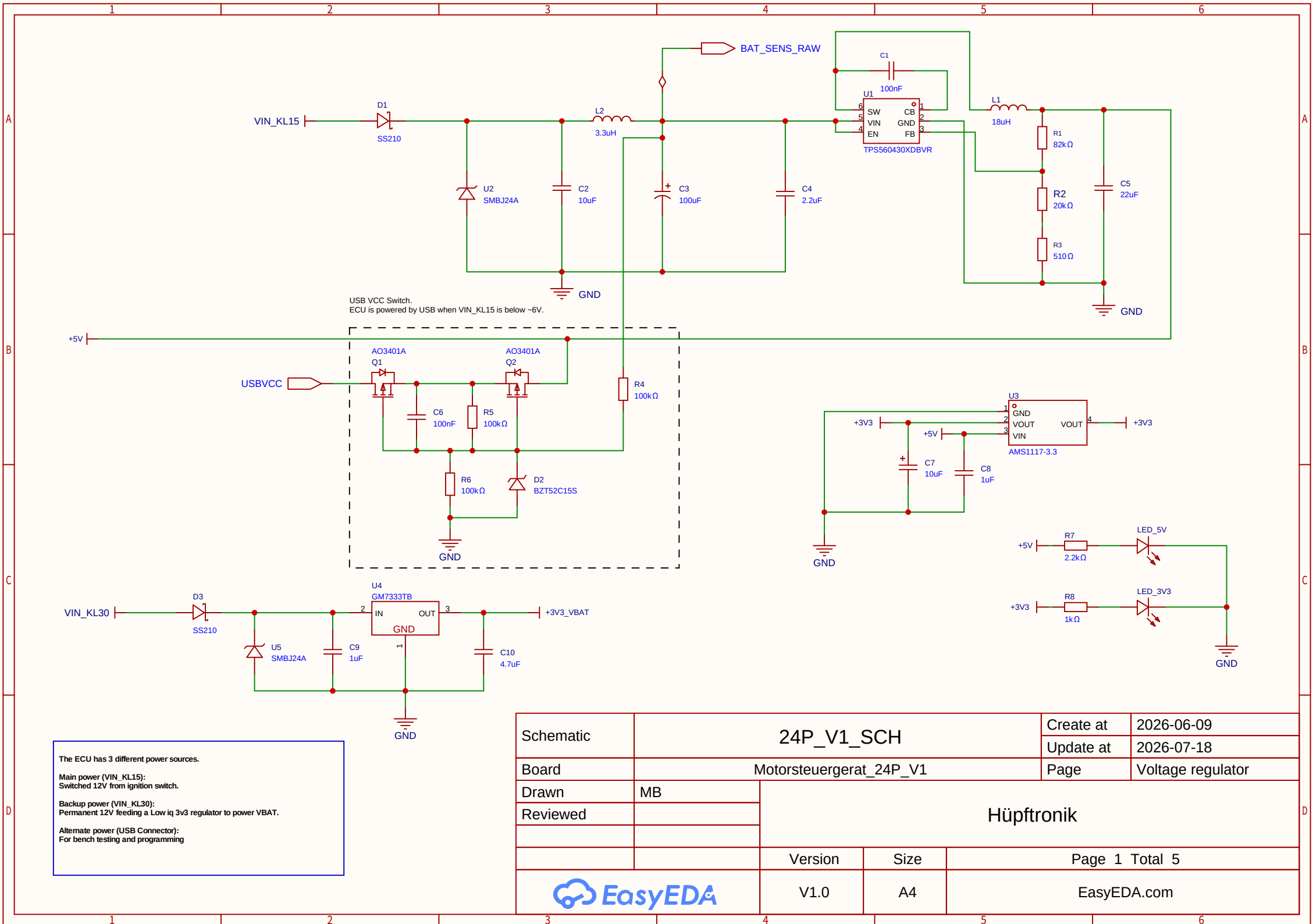




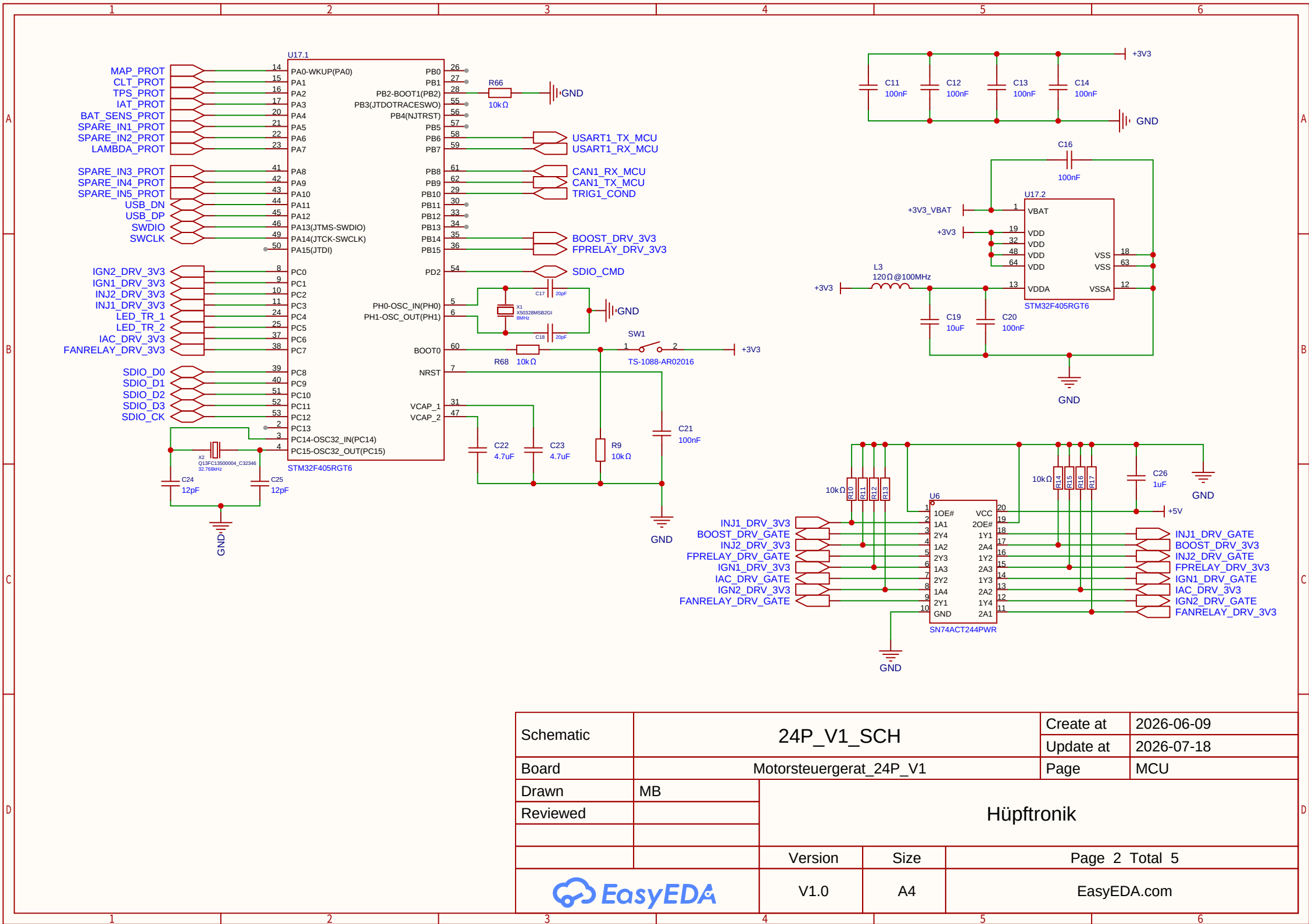
The ECU has 3 different power sources.

Main power (VIN\_KL15):  
Switched 12V from ignition switch.

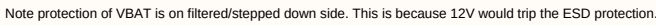
Backup power (VIN\_KL30):  
Permanent 12V feeding a Low iq 3v3 regulator to power VBAT.

Alternate power (USB Connector):  
For bench testing and programming

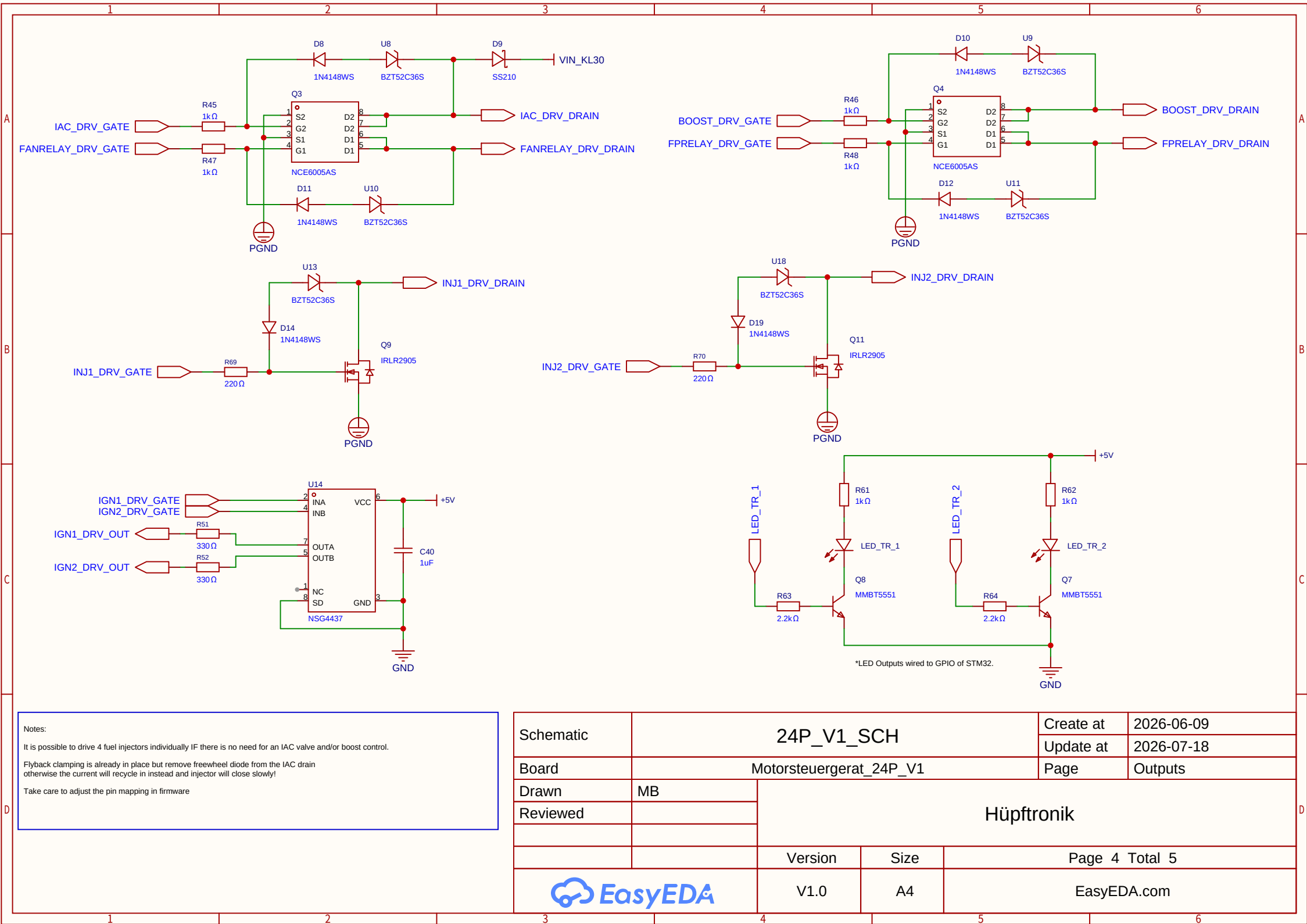
|           |                         |            |      |                |            |
|-----------|-------------------------|------------|------|----------------|------------|
| Schematic | 24P_V1_SCH              |            |      | Create at      | 2026-06-09 |
| Board     | Motorsteuergerat_24P_V1 |            |      | Update at      | 2026-07-18 |
| Drawn     | MB                      | Hüpftronik |      |                |            |
| Reviewed  |                         |            |      |                |            |
|           |                         |            |      |                |            |
|           |                         | Version    | Size | Page 1 Total 5 |            |
| EasyEDA   |                         | V1.0       | A4   | EasyEDA.com    |            |

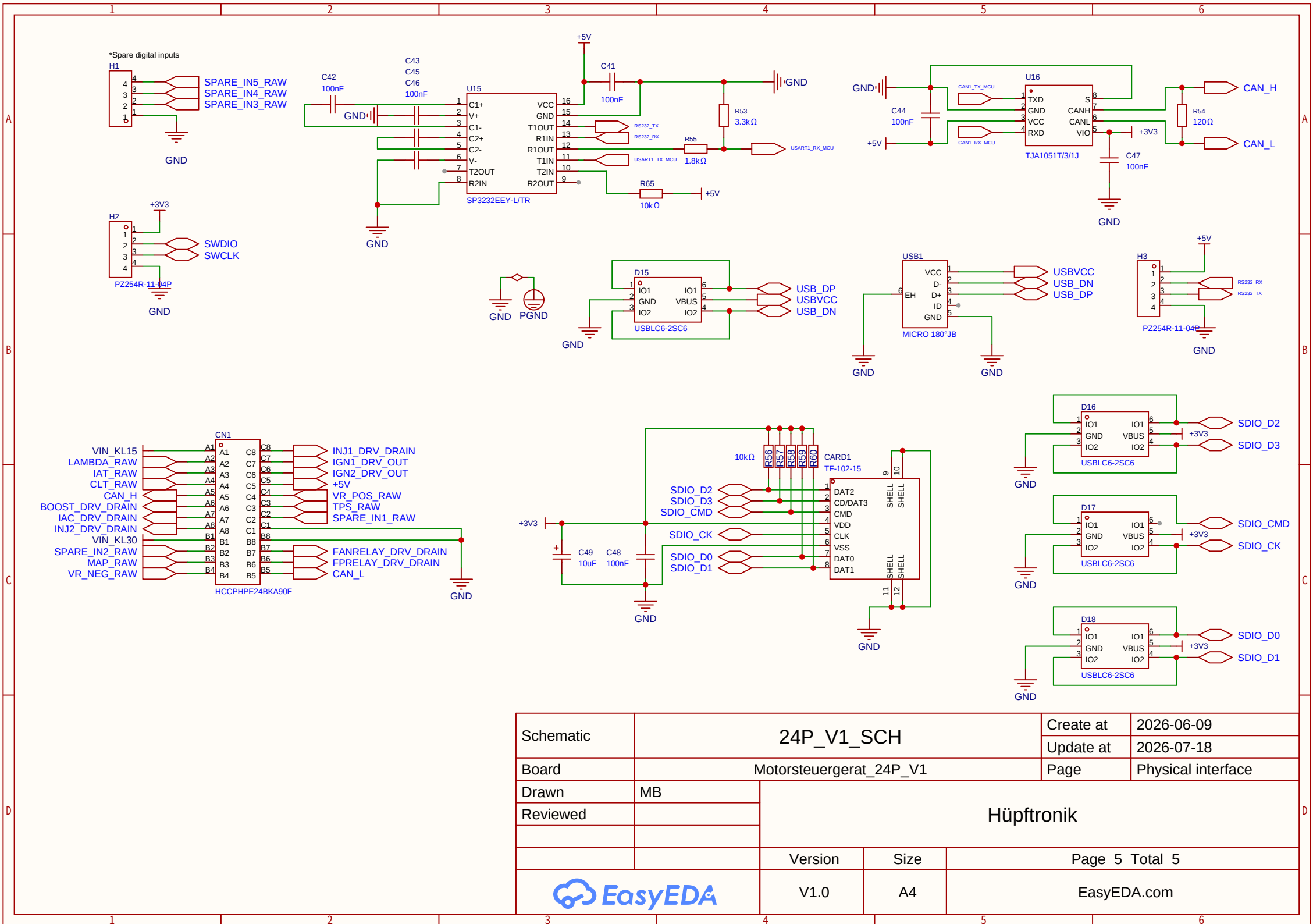


|           |                         |            |      |                |            |
|-----------|-------------------------|------------|------|----------------|------------|
| Schematic | 24P_V1_SCH              |            |      | Create at      | 2026-06-09 |
|           |                         |            |      | Update at      | 2026-07-18 |
| Board     | Motorsteuergerät_24P_V1 |            |      | Page           | MCU        |
| Drawn     | MB                      | Hüpftronik |      |                |            |
| Reviewed  |                         |            |      |                |            |
|           |                         |            |      |                |            |
|           |                         | Version    | Size | Page 2 Total 5 |            |
|           |                         | V1.0       | A4   | EasyEDA.com    |            |



|                                                                                       |                         |            |      |                |            |
|---------------------------------------------------------------------------------------|-------------------------|------------|------|----------------|------------|
| Schematic                                                                             | 24P_V1_SCH              |            |      | Create at      | 2026-06-09 |
|                                                                                       |                         |            |      | Update at      | 2026-07-18 |
| Board                                                                                 | Motorsteuergerat_24P_V1 |            |      | Page           | Inputs     |
| Drawn                                                                                 | MB                      | Hüpftronik |      |                |            |
| Reviewed                                                                              |                         |            |      |                |            |
|                                                                                       |                         |            |      |                |            |
|                                                                                       |                         | Version    | Size | Page 3 Total 5 |            |
|  |                         | V1.0       | A4   | EasyEDA.com    |            |





|           |                         |         |      |                |                    |
|-----------|-------------------------|---------|------|----------------|--------------------|
| Schematic | 24P_V1_SCH              |         |      | Create at      | 2026-06-09         |
|           |                         |         |      | Update at      | 2026-07-18         |
| Board     | Motorsteuergerat_24P_V1 |         |      | Page           | Physical interface |
| Drawn     | MB                      |         |      | Hüpftronik     |                    |
| Reviewed  |                         |         |      |                |                    |
|           |                         | Version | Size | Page 5 Total 5 |                    |
|           |                         | V1.0    | A4   | EasyEDA.com    |                    |